

OOP PROJECT REPORT

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Classes code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace PetsimulatorGUI
{
    class Pet
    {
        public string name;
        public int battery = 100;
        private Mood mood;
        public Pet(string Name, int battery)
        {
            this.name = Name;
            this.battery = battery;
            this.mood = new Mood(battery);
        }
        public Pet(string Name) : this(Name, 100)
        {}
        public string checkmood()
        {
            mood = new Mood(battery);
            return name + " is " + mood.GetMood() + "!";
        }
    }
}
```

```

virtual public string toeat()
{
    battery += 10;
    if (battery > 100)
        battery = 100;
    return name + " " + "is eating now!";
}
virtual public string toplay()
{
    battery -= 10;
    if (battery < 0)
        battery = 0;
    return name + " " + "is playing now!";
}
virtual public string sound()
{
    return name + " " + "is making a sound";
}
}

```

```

class Mood
{
    private int batteryLevel;

    public Mood(int batteryLevel)
    {
        this.batteryLevel = batteryLevel;
    }

    public string GetMood()
    {
        if (batteryLevel > 80)
            return "Energetic";
        else if (batteryLevel > 60)

```

```
        return "Happy";
    else if (batteryLevel > 40)
        return "Okay";
    else if (batteryLevel > 20)
        return "Tired";
    else if (batteryLevel > 5)
        return "Almost Dead";
    else
        return "Dead";
}
}
```

```
class Mammal : Pet
{
    public Mammal(string Name, int battery) : base(Name, battery)
    { }
    public Mammal(string Name) : this(Name, 100)
    { }
    public string sit()
    {
        return name + " is sitting!";
    }
}
```

```
class Dog : Mammal
{
    public Dog(string Name, int battery) : base(Name, battery) { }
    public Dog(string Name) : this(Name, 100)
    { }

    public override string sound()
```

```

{
    return name + " " + "is barking!";
}
public string tail()
{
    if (battery == 100 || battery > 50)
    {
        return name + " is waving its tail!";
    }
    return name + " is too tired to wave its tail.";
}

}

```

```

class Cat : Pet
{
    public Cat(string Name, int battery) : base(Name, battery)
    {}
    public Cat(string Name) : this(Name, 100)
    {}
    public override string sound()
    {
        return name + " " + "is meowing!";
    }
    public string purr()
    {
        if (battery == 100 || battery > 50)
        {
            return name + " " + "is purring!";
        }
        return name + " " + "is too tired now.";
    }
}

```

```
}
```

```
class Bird : Pet
{
    public Bird(string Name, int battery) : base(Name, battery)
    {}
    public Bird(string Name) : this(Name, 100)
    {}
    public override string sound()
    {
        return name + " " + "is singing!";
    }
    public string Fly()
    {
        if (battery == 100 || battery > 50)
        {
            return name + " is flying away!";
        }
        return name + " is too tired now.";
    }
}
```

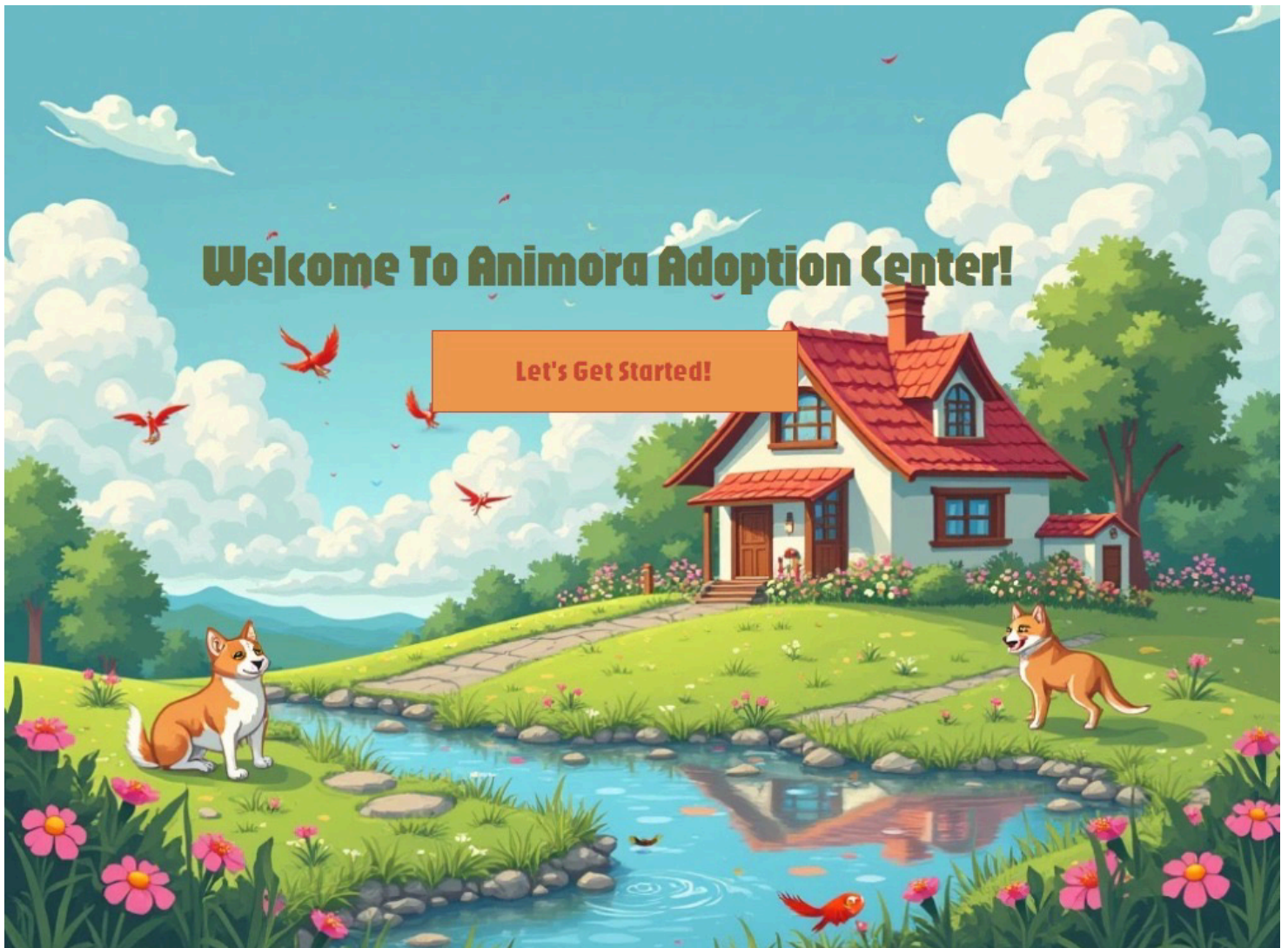
```
class Fish : Pet
{
    public Fish(string name, int battery) : base(name, battery)
    {}
}
```

```
public Fish(string Name) : this(Name, 100)
{
    public string bubbles()
    {
        return name + " is making bubbles!";
    }
}
}
```

Problem statement

adopting a pet and taking care of it with our methods that we made in the program code

FORMS



Username: Animora1

BirthDate: 12/31/2024

Invalid Input



X Error: Name must contain only letters AND birth year must be 2019 or earlier.

Back

OK

Next

CHOOSE YOUR NEW BEST FRIEND!



DOG



CAT



FISH



BIRD

Selection Required



X You must select a character before proceeding.

OK

Back

Next

CHOOSE YOUR PET'S NAME AND AGE!

Pet's Name:

Pet's Age:

Invalid Name



✗ Name must contain only letters.

BACK

NEXT

Form5

Pet Adoption



Back

Confirm



GAME ENDED

Restart